Full Configuration Interaction for Heteronuclear Diatomics in Concatenated Atom-Centred Bases: LiH as a Benchmark*

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We present a Topological LCAO architecture for heteronuclear full configuration interaction (FCI) within the GeoVac discrete graph framework. Two independent atom-centered hydrogenic lattices are concatenated into a bipartite molecular graph via Mulliken cross-nuclear attraction on the diagonal, Slater–Condon two-electron repulsion integrals with exact R^k radial integrals and Gaunt coefficients, and STO-overlap bridges on the off-diagonal; the resulting one- and two-electron Hamiltonian is diagonalized in the full Slater determinant space (4 845 determinants for LiH at $n_{\max} = 2$, where the results below are obtained; the solver also handles the 367 290 determinants of $n_{\max} = 3$). For LiH the balanced Hamiltonian produces a monotonically attractive potential energy surface with no equilibrium geometry; at $n_{\max} = 2$ the curve still lies 0.22 Ha below the separated-atom energy at $R = 8$ bohr, so it does not dissociate to the atoms and no binding energy is extracted. The absence of a minimum is traced to the R-independence of the intra-atom one-electron block: unlike continuous quantum mechanics, each atom's hydrogenic diagonal $-Z^2/(2n^2)$ is fixed at construction and does not respond to inter-nuclear compression; the large-R offset is not diagnosed here. A Fock-weighted kinetic correction, whose *form* is motivated by the energy-shell constraint $p_0^2 = Z^2/n^2$ of Paper 7 but whose overall scale λ is a single fitted parameter (retained as a labeled qualitative diagnostic, not a zero-parameter result), partially restores the equilibrium by concentrating orthogonalization repulsion on compact core orbitals, giving $R_{\text{eq}} \approx 3.7$ bohr at $\lambda = 0.1$ ($n_{\max} = 2$) without repairing the large-R limit. A 29-version diagnostic arc (v0.9.8–v0.9.37) systematically tested and eliminated every competing hypothesis: basis set superposition error as a geometric driver, integral incompleteness, orbital exponent relaxation, the single-n, shared-p_0 SO(4) Sturmian bond sphere approximation, and Hamiltonian imbalance between cross-nuclear and electron repulsion orbital subspaces. In particular, Wigner D-matrix selection rules $D^2_{10,10}(\gamma) = 1$ and $D^2_{00,10}(\gamma) = 0$ are proven as structural results. The excitation-driven Direct CI solver achieves $O(N_{\text{SD}} \times N_{\text{connected}})$ scaling, preserving the $O(V)$ character of the underlying graph Hamiltonian.

I. INTRODUCTION

The GeoVac program works on the discrete label set (n, l, m) of the unit three-sphere S^3, which carries the single-particle quantum-number structure without any continuous spatial integration [1–3]. Prior work established the exact correspondence at the level of labels for single-center atomic systems—on them the S^3 Laplacian is the diagonal $SO(4)$ Casimir, whose spectrum Fock's projection maps to the hydrogenic one—with $O(V)$ scaling; the graph Laplacian itself does not converge to the S^3 operator (Paper 7, corrected 2026-09-25). Extending this framework to multi-center molecules, however, presents a fundamental geometric challenge: two or more atomic lattices, each living on its own S^3 conformal manifold, must be bridged into a single molecular graph without resorting to the continuous $\mathbb{R}^3$ integration that the discrete premise is designed to replace.

In this work we introduce the *Topological LCAO* architecture, which solves the multi-center problem by concatenating atomic subgraphs into a bipartite molecular graph connected by discrete off-diagonal bridges. The resulting block Hamiltonian is fed into an $O(V)$ excitation-driven Full Configuration Interaction solver that operates entirely in Slater determinant space. The headline result is negative: for LiH the balanced Hamiltonian gives a monotonically attractive potential energy surface with no equilibrium geometry, revealing a fundamental limitation of the discrete LCAO framework: the absence of R-dependent kinetic repulsion. A Fock-weighted kinetic correction, whose form follows Paper 7 [3] but whose scale λ is fitted, partially restores an equilibrium.

LiH is the minimal four-electron heteronuclear diatomic and therefore the ideal first benchmark for the Topological LCAO framework. Its four electrons permit exact FCI treatment without truncation of the many-body expansion, while the $Z_A = 3$, $Z_B = 1$ asymmetry breaks the inversion symmetry exploited by homonuclear codes and exposes every heteronuclear coupling pathway. The experimental targets are well-established: $D_e = 0.092$ Ha and $R_{\text{eq}} = 3.015$ bohr [7]. This paper formally extends the companion computational framework described in Paper 8 [4].

The paper is organized as follows. Section II develops the bipartite graph construction and the $O(V)$ Direct CI solver. Section III presents the LiH benchmark results, including the counterpoise-corrected potential energy surface and the Hamiltonian balance requirement. Section IV documents the 29-version diagnostic arc that systematically falsified every hypothesis for the observed behavior—integral incompleteness, orbital relaxation, and Sturmian geometries. Section V identifies

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the root cause as missing R-dependent kinetic repulsion and presents a Fock-weighted correction whose form follows Paper 7 and whose scale is fitted. Section VI summarizes and outlines next steps.

II. TOPOLOGICAL LCAO ARCHITECTURE

A. Subgraph Concatenation

The `MolecularLatticeIndex` instantiates two independent atomic subgraphs: a lithium lattice with nuclear charge $Z_A = 3$ and a hydrogen lattice with $Z_B = 1$. Each subgraph is a scale-invariant hydrogenic lattice defined on its own S^3 conformal manifold [3], retaining its atom-centered conformal structure; the hydrogenic energies $\varepsilon_n = -Z^2/(2n^2)$ are assigned to the labels n through Fock's energy shell $p_0^2 = -2E$ (the bare graph Laplacian is positive-semidefinite and does not converge to Δ_{S^3}, so these negative energies are neither its eigenvalues nor a continuum image of its spectrum). At $n_{\max} = 3$, the lithium subgraph contributes $\sum_{n=1}^{3} n^2 = 14$ spatial orbitals and the hydrogen subgraph contributes another 14, yielding a combined molecular basis of 28 spatial orbitals (56 spin-orbitals).

The concatenation is a direct sum at the graph level: the adjacency matrix of the molecular graph is block-diagonal in the two atomic subgraphs prior to bridging. No spatial overlap integral is computed between the subgraphs at this stage; the topological identity of each atomic lattice is preserved exactly.

B. Topological Bridging and the Hamiltonian

Bonding emerges from a rank-update concatenation that introduces off-diagonal coupling between the two atomic blocks. The one-electron molecular Hamiltonian takes the block form

$$\mathbf{H}_1 = \begin{pmatrix} \mathbf{H}_A & \mathbf{V}_{AB} \\ \mathbf{V}_{AB}^\dagger & \mathbf{H}_B \end{pmatrix}, \tag{1}$$

where $\mathbf{H}_A$ and $\mathbf{H}_B$ are the isolated atomic Hamiltonians and $\mathbf{V}_{AB}$ encodes the inter-atomic coupling.

The diagonal blocks carry the hydrogenic eigenvalues $\varepsilon_n = -Z^2/(2n^2)$ (the energy-shell map of the labels n, not graph-Laplacian eigenvalues and not an image of the graph spectrum) supplemented by the Mulliken cross-nuclear attraction: each orbital $\phi_i^{(A)}$ centered on atom A experiences the potential of nucleus B via the diagonal matrix element

$$\langle \phi_i^{(A)} | -Z_B/r_B | \phi_i^{(A)} \rangle, \tag{2}$$

evaluated analytically from the hydrogenic Fourier representation. Off-diagonal elements of $\mathbf{V}_{AB}$ are STO-overlap bridges weighted by conformal factors derived from the stereographic projection [5].

The two-electron repulsion V_{ee} is constructed via Slater–Condon rules with two classes of integrals. Same-atom Coulomb and exchange integrals use exact Slater R^k radial integrals with Gaunt angular coefficients, validated against the known analytical values $F^0(1s,1s) = 5Z/8$, $F^0(1s,2s) = 17Z/81$, and $F^0(2s,2s) = 77Z/512$. Cross-atom Coulomb repulsion is evaluated via multipole expansion, with exchange restricted to s-orbital pairs where the overlap is non-negligible.

C. $O(V)$ Direct Configuration Interaction

Distributing four electrons across 56 spin-orbitals generates $\binom{56}{4} = 367\,290$ Slater determinants. A naïve pairwise construction of the CI Hamiltonian matrix would require $O(N_{\mathrm{SD}}^2)$ evaluations—over 10^{11} matrix element computations—rendering exact diagonalization impractical.

We instead adopt the excitation-driven Direct CI algorithm of Knowles and Handy [8], which constructs the Hamiltonian action $\sigma = \mathbf{H}|\Psi\rangle$ by traversing only the nonzero edges generated by valid single and double excitations from each determinant. The computational cost scales as $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$, where $N_{\mathrm{connected}}$ is the average number of determinants connected to each reference by at most a double excitation. For the LiH basis at $n_{\max} = 3$, this reduces wall time from hours to approximately 8 seconds when the Slater–Condon assembly loops are JIT-compiled via Numba [9] (approximately 75 seconds in pure Python).

The one-electron integrals h_{pq} and two-electron integrals $(pq|rs)$ are stored as dense arrays (56 spin-orbitals yield lookup tables of order 56^2 and 28^4, both fitting comfortably in memory), eliminating the $\sim 24\,\mu$s per-access overhead of `scipy.sparse` index validation that dominated earlier implementations.

D. Counterpoise Correction Framework

Basis set superposition error (BSSE) is an inherent artifact of any atom-centered expansion: at finite basis size, each atom can variationally borrow orbitals from its partner, artificially lowering the molecular energy below the true complete-basis-set limit. We implement the Boys–Bernardi counterpoise (CP) correction [6] natively within the Topological LCAO framework.

Ghost fragments are constructed by preserving the full bipartite topology of the molecular graph while zeroing the nuclear charge and electron population of the partner center. Specifically, a ghost-B calculation retains all 28 spatial orbitals but sets $Z_B = 0$, eliminating all bridges, cross-nuclear attraction, and cross-atom V_{ee} involving center B. The fragment energy E_A^{mol} is then computed in the identical topological basis space as the supermolecule, guaranteeing a consistent subtraction:

$$D_e^{\mathrm{CP}} = E_A^{\mathrm{mol}} + E_B^{\mathrm{mol}} - E_{AB}(R_{\mathrm{eq}}). \tag{3}$$

III. LIH BENCHMARK RESULTS

A. Computational Parameters

The molecular graph is assembled from two atomic subgraphs at $n_{\max} = 3$: a lithium lattice ($Z_A = 3$, 14 spatial orbitals) and a hydrogen lattice ($Z_B = 1$, 14 spatial orbitals), yielding a combined basis of 28 spatial orbitals and 56 spin-orbitals. The full four-electron Slater determinant space contains $\binom{56}{4} = 367\,290$ configurations. The exact ground-state eigenvalue is obtained from the excitation-driven Direct CI solver described in Sec. II C, with wall time of approximately 8 seconds on a single CPU core (Numba-accelerated assembly; approximately 75 seconds without JIT compilation). The energetics below are reported at $n_{\max} = 2$ (5 spatial orbitals per atom, 20 spin-orbitals, 4 845 determinants), the size at which the physical one-electron Hamiltonian has been run; the $n_{\max} = 3$ figures above describe the method and its cost.

B. Energetic Accuracy and Basis Set Superposition

Table I gives the LiH potential energy surface at $n_{\max} = 2$ with the physical one-electron Hamiltonian: the hydrogenic diagonal plus cross-nuclear attraction, with the STO-overlap bridges as the only off-diagonal one-electron terms. The separated-atom reference (each atom in its own basis) is $E_{\mathrm{Li}} + E_{\mathrm{H}} = -7.601$ Ha, and the Boys–Bernardi counterpoise correction gives $\Delta E_{\mathrm{BSSE}} = 0.088$ Ha, the same at every R from 1.5 to 8 bohr (the ghost fragments carry no R-dependent term). The surface does not return to the separated-atom energy at large R (Sec. III C), so the energy difference to the atomic fragments at $R = 3.015$ bohr is not a dissociation energy and is not compared with experiment.

R (bohr)	E (Ha)
1.5	−8.3793
2.5	−8.0592
3.015	−7.9899
4.0	−7.9312
5.0	−7.8877
8.0	−7.8258
separated atoms	−7.601

TABLE I. LiH potential energy surface at $n_{\max} = 2$ (balanced `exact+True` configuration, physical one-electron Hamiltonian). The energy falls monotonically as R decreases, with no minimum, and at $R = 8$ bohr it lies 0.22 Ha below the separated-atom energy.

a. Hamiltonian balance requirement. The cross-nuclear attraction and electron repulsion must operate on matched orbital subspaces. Two balanced configurations—`exact` cross-nuclear with full cross-atom V_{ee} (`exact+True`), and `fourier` cross-nuclear with s-only V_{ee} (`fourier+s_only`)—produce nearly identical molecular energies across the PES. An *imbalanced* configuration (`exact+s_only`) artificially overbinds, as the p and d orbitals receive cross-nuclear attraction without the compensating electron repulsion screening. The results presented here are from the balanced `exact+True` Hamiltonian.

C. Absence of Equilibrium Geometry

The balanced configurations (`exact+True` and `fourier+s_only`) both yield monotonically attractive PES curves with no equilibrium geometry.

With the physical one-electron Hamiltonian (Table I) the energy falls from −7.826 Ha at $R = 8.0$ bohr to −8.379 Ha at $R = 1.5$ bohr, the slope diminishing at large R but never reversing sign, and at $R = 8$ bohr the curve still lies 0.22 Ha below the separated-atom energy. At no point does a repulsive wall balance the attractive cross-nuclear and exchange interactions: the system is variationally attracted to short bond lengths at all R.

This monotonic behavior is traced to a structural property of the Topological LCAO: the one-electron block of each atom is *completely R-independent*. In continuous quantum mechanics, bringing two atoms together forces their electrons into higher-momentum orthogonal states, creating the kinetic repulsion that establishes the equilibrium bond length. In the discrete framework, each atom's one-electron block—the hydrogenic diagonal $-Z^2/(2n^2)$, which carries that atom's kinetic energy—is fixed at construction time and does not respond to the approach of a second center: the atom-centered orbitals are treated as orthonormal across the two centers, so no orthogonalization kinetic energy arises.

The only R-dependent off-diagonal one-electron terms are the bridge edges connecting the two subgraphs, and they are small: the maximum bridge matrix element at $R = 3.015$ bohr is 0.017 Ha, and the bridges produce no repulsive wall.

This absence of equilibrium geometry motivates both the systematic diagnostic arc in the following section and the kinetic repulsion analysis in Sec. V.

IV. SYSTEMATIC DIAGNOSTIC ARC

The initial implementation (v0.9.8) exhibited heavy overbinding ($D_e^{\mathrm{raw}} = 0.205$ Ha, more than twice the experimental value) and a 17% geometric contraction of the equilibrium bond length. Over the course of 29 controlled experiments spanning versions v0.9.8 through v0.9.36, every plausible hypothesis for these discrepancies was systematically tested. The experiments are grouped below by hypothesis class; the complete version history is tabulated in Appendix A.

A. Hypothesis 1: Basis Set Superposition Error (v0.9.8–v0.9.11)

The first hypothesis attributed both the energetic overestimation and the geometric contraction to BSSE: electrons borrowing orbitals from the partner center would artificially lower the molecular energy, inflating D_e and potentially distorting the PES minimum.

The Boys–Bernardi counterpoise correction (Sec. II D) quantified the superposition error ($\Delta E_{\mathrm{BSSE}} = 0.088$ Ha at $n_{\max} = 2$ with the physical one-electron Hamiltonian, Table I). BSSE does not address the absence of an equilibrium geometry, which has a distinct root cause (Sec. V).

B. Hypothesis 2: Integral Incompleteness (v0.9.12–v0.9.13, v0.9.35)

The second hypothesis class attributed the geometric contraction to missing or approximate molecular integrals: either the Mulliken cross-nuclear attraction overestimates the short-range potential, or the restriction of cross-atom V_{ee} to s-orbital pairs neglects significant contributions from higher angular momentum channels.

Exact cross-nuclear attraction matrix elements, computed analytically from the hydrogenic Fourier representation and validated against the shell theorem, deviate by less than 2 mHa from the Mulliken diagonal approximation at $n_{\max} = 2$ ($R = 2.5$–5.0 bohr), confirming that the Mulliken scheme is close to exact for this system.

With the Fourier cross-nuclear diagonal, extending the cross-atom V_{ee} to all (n, l) pairs—392 Coulomb integrals at $n_{\max} = 3$ compared to 18 for the s-only restriction—shifts the total energy by only $+0.3$ to $+0.4$ mHa at $n_{\max} = 2$. This negligible shift reflects the fact that the LiH ground state is overwhelmingly s-orbital dominated: the $1s^2 2s^2$ configuration of lithium and the $1s$ of hydrogen leave higher angular momentum channels essentially unoccupied.

Integral incompleteness is conclusively falsified as a mechanism for the geometric contraction.

C. Hypothesis 3: Orbital Exponent Relaxation (v0.9.36)

The third hypothesis proposed that fixing atomic orbital exponents at their free-atom values forces artificial proximity: if electrons cannot polarize toward the bonding region by adjusting their radial extent, the system compensates by contracting R_{eq}.

A variational orbital exponent parameter ζ_B was introduced for the hydrogen center, allowing the effective nuclear charge $Z_{\mathrm{eff}} = \zeta_B Z_B$ to optimize the radial extent of all hydrogen-centered orbitals. At $n_{\max} = 2$ with the physical one-electron Hamiltonian (scan step 0.1), the variational minimum occurs at $\zeta_B \approx 1.3$ at $R = 3.015$ bohr, relaxing toward unity at large R ($\zeta_B \approx 1.1$ at $R = 5$ and 8 bohr).

The relaxation energy is approximately -0.31 Ha, but it is *uniform across the bonding region*: the energy lowering at $R = 2.5$ bohr and $R = 3.015$ bohr differs by only 0.004 Ha. Consequently the relaxed PES remains monotonically attractive, with no minimum. Moreover, the energy below the counterpoise ghost fragments at $R = 3.015$ bohr grows by a factor of 2.2 (from 0.300 to 0.652 Ha), demonstrating that the variational freedom is exploited for unphysical basis compression rather than physical polarization.

Orbital exponent relaxation is eliminated as a mechanism for geometric contraction.

D. Hypothesis 4: Bond Sphere and Sturmian Basis (v0.9.15–v0.9.34)

The fourth and most extensive hypothesis class proposed that the flat-space LCAO concatenation is fundamentally incompatible with the S^3 topology, and that a proper molecular treatment requires Sturmian basis functions defined on a single *bond sphere*—an S^3 manifold with two weighted poles—connected by SO(4) Wigner D-matrix elements.

This program was pursued across 20 versions (v0.9.15–v0.9.34), progressing from the initial SO(4) Wigner D-matrix implementation (44 tests, v0.9.15) through geometric-mean coupling (v0.9.16, unbound), Shibuya–Wulfman off-diagonal nuclear attraction (v0.9.17, unbound), hybrid Fourier–SW schemes (v0.9.18, 55% overbinding), atom-centered Sturmian variants (v0.9.20–v0.9.28, all unbound or massively overbinding), prolate spheroidal MO projections (v0.9.29–v0.9.30, unphysical), full MO Sturmian FCI (v0.9.31–v0.9.33, 12–25% error), and dual-p_0 extensions (v0.9.34, 25% error with p_0 values collapsing to a single geometric mean).

The definitive negative result is the structural theorem proven in companion Paper 8 [4] (Theorem 4): in a Sturmian basis sharing a single momentum parameter p_0, treated with Paper 8's single-n rotation approximation, the one-electron Hamiltonian and overlap obey the two-term identity $H_{ij} = -\frac{1}{2}p_0^2 S_{ij} + (1 - \beta_j)V_{ij}$ and share a single orthogonal $SO(4)$ Wigner-D congruence $D^{(n)}(\gamma(R))$ (so H is block-, not scalar-, proportional to S). The generalized eigenvalues are therefore independent of the bond angle γ, hence of R, yielding pure $1/R$ nuclear repulsion with no electronic binding. Within that approximation, recovering R-dependent binding energies requires solving the continuous two-center prolate spheroidal PDE for the spectrum $\beta_k(R)$, reintroducing exactly the continuous spatial geometry that the discrete graph framework is designed to eliminate. The Bond Sphere Sturmian alternative is therefore falsified *in this single-n, shared-p_0 form*. The scope matters: the genuine cross-n Coulomb–Sturmian overlap couples n and escapes the theorem, so binding is not excluded for it (Paper 8,

Remark). What is ruled out is this approximation, not the discrete framework.

a. Cross-n hybridization (v0.9.37). A further hypothesis attributed the geometric error to the D-matrix's block-diagonality in n: the single-n D-matrix approximation of Paper 8 connects only same-n orbitals (Li $1s \leftrightarrow$ H $1s$, Li $2s \leftrightarrow$ H $2s$), giving zero off-diagonal coupling for the chemically relevant Li $2s \leftrightarrow$ H $1s$ pair, which the full cross-n Shibuya–Wulfman matrix does couple [4]. Overlap-based bridges using the Shibuya–Wulfman energy scale $-(Z_{\text{eff}}/p_0)\sin\gamma$ were implemented, producing Li $2s$–H $1s$ coupling of 0.052 Ha (40% of the same-n D-matrix coupling). However, the PES worsened: the cross-n coupling adds binding at short R, where the overlap is largest. Cross-shell hybridization is eliminated as a mechanism for the geometry error.

b. Level 4 locked-core (v0.9.37). The Level 4 molecule-frame hyperspherical solver (Paper 15) was extended to LiH by locking the Li $1s^2$ core and treating the two active electrons in hyperspherical coordinates with effective charges Z_{eff}. Four core models were tested (bare Z, integer screening, Clementi–Raimondi, and Core Hartree with penetration); all produce PES that collapse to short bond lengths ($R_{\text{eq}} \lesssim 2.5$ bohr) with $D_e > 12$ eV. The failure has the same root cause: the core-to-H-nucleus attraction $V_{\text{cross}} = -2Z_B\langle 1s_Z^2|1/r_B|1s_Z^2\rangle$ behaves as $-2/R$ at large R but saturates exponentially at short R ($\exp(-6R)$ decay for Li $1s$), providing insufficient repulsion to establish a minimum at $R \approx 3$ bohr.

Having falsified BSSE as a geometric driver, eliminated integral incompleteness, ruled out orbital exponent relaxation, shown that the single-n, shared-p_0 Sturmian approximation cannot bind (Paper 8's theorem, which does not exclude the genuine cross-n basis), and excluded cross-n hybridization and locked-core hyperspherical approaches, we trace the absence of an equilibrium geometry in the balanced Hamiltonian to the R-independence of the intra-atom one-electron block, analyzed in the following section.

V. MISSING KINETIC REPULSION AND THE FOCK-WEIGHTED CORRECTION

A. The Kinetic Diagnostic

Direct examination confirms that the intra-atom one-electron block (the hydrogenic diagonal $-Z^2/(2n^2)$ within each atomic subgraph) is identically R-independent: it is fixed by the isolated-atom labels and never modified as R changes. Bridge edges contribute a small R-dependent off-diagonal term, but the maximum bridge element at $R = 3.015$ bohr is only 0.017 Ha, and it produces no repulsive wall.

This is a structural limitation of the Topological LCAO architecture: the concatenated basis, treated as orthonormal across the two centers, has no mechanism to represent the orthogonalization kinetic energy that arises when atomic orbitals overlap in continuous quantum mechanics.

B. The Fock-Weighted Kinetic Correction

The energy-shell constraint of Paper 7—$p_0^2 = -2E = Z^2/n^2$ for the n-th shell—provides a natural momentum scale $p_0 = Z/n$ for each orbital. We exploit this to construct a physically motivated kinetic correction.

For each orbital ϕ_a on atom A, the Fock-weighted overlap kinetic correction adds to the diagonal of $\mathbf{H}_1$:

$$\Delta h_{aa} = \lambda \sum_{b \in B} S_{ab}^2 \left(\frac{Z_A}{n_a}\right)^2 \left(\frac{Z_B}{n_b}\right)^2, \tag{4}$$

where $S_{ab} = \langle \phi_a | \phi_b \rangle$ is the STO overlap integral between s-orbitals on different centers, and λ is a single adjustable parameter. The Fock weights suppress diffuse orbitals (high n) and concentrate the kinetic penalty on compact core orbitals whose overlap decays rapidly with R.

The physical content of Eq. (4) is that the cost of overlapping two orbitals scales with the product of their squared shell momenta $p_0^2 = Z^2/n^2$. Core–core pairs (Li $1s$–H $1s$) carry weight $9 \times 1 = 9$, while diffuse–diffuse pairs (Li $3s$–H $3s$) carry weight $1 \times 0.11 = 0.11$—an $81\times$ discrimination.

The R-discrimination (ratio of correction at $R = 1.5$ to $R = 5.0$ bohr) improves from $3.0\times$ with uniform weighting to $9.3\times$ with Fock weighting. For the Li $1s$ orbital alone, the ratio improves from $19\times$ to $74\times$.

C. Results of the Fock-Weighted Correction

Table II summarizes the λ-scan at $n_{\max} = 2$ with the physical one-electron Hamiltonian ($R = 1.5$–8.0 bohr, step 0.5 bohr near the minimum).

λ	Equilibrium?	R_{eq} (bohr)	$E(8.0) - E(R_{\text{eq}})$ (Ha)
0.00	No	—	—
0.02	No	—	—
0.05	No	—	—
0.10	**Yes**	**~ 3.7**	**0.092**
0.20	Yes	~ 4.1	0.075
0.50	Yes	~ 4.7	0.045

TABLE II. Fock-weighted kinetic correction scan at $n_{\max} = 2$ (λ fitted; R_{eq} from a three-point parabola about the lowest grid point). For $\lambda \geq 0.1$ a potential energy minimum appears and moves outward as λ grows; the last column is the depth below the $R = 8$ bohr point, not a dissociation energy.

For $\lambda \geq 0.1$ the Fock-weighted correction creates a potential energy minimum with a repulsive inner wall, at $R_{\text{eq}} \approx 3.7$ bohr for $\lambda = 0.1$, moving to 4.1 and 4.7 bohr at $\lambda = 0.2$ and 0.5. This is a qualitative success—the first

equilibrium geometry produced by any balanced Hamiltonian configuration—but it does not repair the large-R limit: for every λ the curve at $R = 8$ bohr still lies 0.22 Ha below the separated-atom energy, so no dissociation energy is obtained, and no scanned value of λ places the minimum at the experimental 3.015 bohr.

D. Remaining Limitations

The remaining discrepancy ($R_{\mathrm{eq}} \approx 3.7$ vs. experiment 3.015 bohr at $\lambda = 0.1$, and no return to the separated atoms) likely reflects the limitation of a purely diagonal correction. In standard quantum chemistry, kinetic repulsion includes both diagonal (orbital compression) and off-diagonal (orthogonalization) contributions from the full kinetic energy matrix $T_{AB} = \langle\phi_A| - \frac{1}{2}\nabla^2|\phi_B\rangle$. The Fock-weighted correction captures only the diagonal component.

The BSSE of 0.088 Ha at $n_{\max} = 2$ is comparable to the experimental dissociation energy of 0.092 Ha, confirming significant basis deficiency. However, since the balanced PES has no minimum, BSSE is not the mechanism for geometric contraction—it is a separate issue of basis completeness that CP correction addresses for vertical binding energies.

VI. CONCLUSION

We have presented a heteronuclear full configuration interaction calculation in concatenated atom-centered hydrogenic bases on the S^3 label set. The Topological LCAO architecture concatenates atom-centered hydrogenic lattices via Mulliken cross-nuclear attraction, Slater–Condon two-electron integrals, and STO-overlap bridges, and diagonalizes the resulting Hamiltonian in the full Slater-determinant space using excitation-driven Direct CI (4 845 determinants at $n_{\max} = 2$, where the LiH results are obtained; the solver also handles the 367 290 determinants of $n_{\max} = 3$).

For LiH the balanced Hamiltonian produces a monotonically attractive PES with no equilibrium geometry; at $n_{\max} = 2$ it still lies 0.22 Ha below the separated-atom energy at $R = 8$ bohr. The absence of a minimum is traced to the R-independence of each atom's one-electron block, a structural property rather than basis truncation; the large-R offset is not diagnosed here.

A 29-version diagnostic arc (v0.9.8–v0.9.37) systematically falsified every competing hypothesis: basis set superposition error as a geometric driver, integral incompleteness, orbital exponent relaxation, and the single-n, shared-p_0 SO(4) Sturmian Bond Sphere approximation. The missing kinetic repulsion was identified as a structural limitation of the LCAO concatenation, and a Fock-weighted kinetic correction, whose form follows Paper 7's energy-shell constraint $p_0^2 = Z^2/n^2$ and whose scale λ is fitted, partially restores the equilibrium ($R_{\mathrm{eq}} \approx 3.7$ bohr at $\lambda = 0.1$, $n_{\max} = 2$) without repairing the large-R limit.

The $O(V)$ scaling of the underlying graph Hamiltonian is preserved throughout, with the Direct CI solver achieving $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$ assembly cost. The natural next steps are:

1. Extending the Fock-weighted correction to include off-diagonal kinetic matrix elements between atomic centers, which would provide the missing intermediate-R binding;

2. Increasing $n_{\max}$ to assess basis convergence of the potential energy surface and its large-R limit.

3. **Prolate spheroidal lattice:** The fundamental limitation identified above—that the LCAO one-electron blocks provide no R-dependent kinetic repulsion—arises because the molecular graph is constructed as a concatenation of atomic S^3 subgraphs whose one-electron blocks are fixed at construction time. This limitation is resolved by recognizing that the natural geometry for two-center systems is the prolate spheroid, not the concatenated S^3. In prolate spheroidal coordinates (ξ, η, φ), the two-center Coulomb problem separates exactly, and working in the label set and channel structure of these coordinates yields a molecular lattice with R-dependence built into the metric. For H_2^+, this approach achieves $R_{\mathrm{eq}} = 2.001$ bohr (0.21% error) at the finite-difference level, while the spectral Laguerre radial solver reaches machine precision, both with zero fitted parameters [10]. The LCAO framework documented in this paper remains valuable for understanding the anatomy of molecular binding in a discrete framework, and the diagnostic arc (Table III) provides a systematic record of approaches that fail and why.

The Fock projection's energy-shell scale $p_0^2 = Z^2/n^2$ supplies a qualitatively useful weighting for the kinetic repulsion, with its overall scale λ fitted.

Appendix A: Version History

Table III documents the complete diagnostic arc from v0.9.11 (the balanced LCAO baseline) through v0.9.38, including architecture, results, and root cause classification for each version.

[1] J. Loutey, "Counting Angular Momentum Labels by Isotropic Packing," GeoVac Paper 0 (2026).

[2] J. Loutey, "The Geometric Atom: the Hydrogen State Space as a Label Lattice," GeoVac Paper 1 (2026).

Version	Architecture	D_e^{CP} (Ha)	R_{eq} (bohr)	Status	Root Cause
v0.9.11	LCAO baseline (balanced `exact+True`)	—	No eq.	Baseline	No large-R return to atoms
v0.9.12	Löwdin orthogonalization	—	—	Fail	Breaks variational principle
v0.9.13	Mulliken exchange K	—	$\sim$2.5	—	K overcorrects D_e
v0.9.15	SO(4) Wigner D-matrix	—	—	Infrastructure	44/44 tests, no production
v0.9.16	D-matrix cross-nuclear (geom. mean)	—	—	Unbound	Coupling too weak
v0.9.17	Shibuya–Wulfman off-diagonal	—	—	Unbound	SW coupling too weak
v0.9.18	Hybrid (Fourier diag + SW off-diag)	0.143	$<$2.0	55% err	Dissociation limit wrong
v0.9.20	Sturmian diagonal	—	—	Unbound	$1s$ diagonal positive
v0.9.21	p_0-consistent D-matrix + self-loop	—	—	Unbound	Compactness overbinding
v0.9.22	Atom-dependent p_0	3.777	$\sim$2.0	3.8$\times$ overbind	Scales do not decouple
v0.9.23	Shell-radius attenuation	—	—	Negligible	Correction $<$ 1 mHa
v0.9.24	Energy decomposition diagnostic	—	—	Tool	Identifies Fourier diagonal
v0.9.25	Ohno–Klopman cross-atom V_{ee}	—	—	Negligible	Cross-atom J irrelevant
v0.9.26	Angular-weighted V_{ee}	—	—	Negligible	$\Delta E \sim 0.4$ mHa
v0.9.27	SO(4) CG cross-center V_{ee}	—	—	Unbound	$J_{\mathrm{cross}} > J_{\mathrm{same}}$
v0.9.28	Bond sphere node weights	—	—	Unbound	β_H 46% short
v0.9.29	Prolate spheroidal MO betas	—	—	Unbound	LiH mapping ad hoc
v0.9.30	MO-to-atom projection	—	—	Unphysical	$E = -42.5$ Ha (5.3$\times$)
v0.9.31	MO Sturmian FCI (full)	—	—	Unbound/overbinding	Shared $SO(4)$ congruence, eigenvalues γ-ind
v0.9.32	MO Sturmian FCI (variant)	—	—	Unbound/overbinding	Shared $SO(4)$ congruence, eigenvalues γ-ind
v0.9.33	MO Sturmian FCI (variant)	—	—	Unbound/overbinding	Shared $SO(4)$ congruence, eigenvalues γ-ind
v0.9.34	Dual-p_0 extension	—	—	Unbound/overbinding	Shared $SO(4)$ congruence, eigenvalues γ-ind
v0.9.35	All-l cross-atom V_{ee}	—	No eq.	$\Delta E = 0.4$ mHa	Negligible correction
v0.9.36	Orbital exponent relaxation	—	No eq.	Fail	Uniform PES shift
v0.9.37	Hamiltonian balance diagnostic	—	No eq.	Corrected	Imbalance identified
v0.9.38	Numba-accelerated assembly	—	No eq.	738$\times$ speedup	Python loop overhead eliminated

TABLE III. Diagnostic arc version history (v0.9.11–v0.9.38). The LCAO baseline (v0.9.11, balanced `exact+True`) lacks an equilibrium geometry due to missing kinetic repulsion and does not return to the separated-atom energy at large R (Table I).

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